



University of Debrecen, Faculty of Informatics

Logic in computer science

First-order logic

(predicate logic)

Syntax

November 5, 2022

The first-order language of classical logic

An ordered 5-tuple $\mathcal{L}^{(1)} = \langle LC, Var, Con, Term, Form \rangle$ defines the language of the first-order (predicate) logic, where

1. $LC = \{ \neg, \supset, \wedge, \vee, \equiv, (,), =, \exists, \forall, . \}$ is the set of *logical constants*,
2. Var is the countable infinite set of variables,
3. $Con \neq \emptyset$; is the set of *non-logical constants* such that $LC \cap Con = \emptyset$, and

$$Con = \bigcup_{i=0}^{\infty} (\mathcal{P}(i) \cup \mathcal{F}(i));$$

4. $Term$ is the set of *terms*;
5. $Form$ is the set of *formulae*;

where LC , Var , $\mathcal{P}(i)$ and $\mathcal{F}(i)$ are pairwise disjoint.

The LC , Var and Con sets contains the letters of the alphabet, while the terms and formulas as sequences of letters are the words of the first-order language.

The first-order language – logical constants

The first-order logical constants are

- the unary connective:
 - $\neg$ negation sign
- the binary connectives:
 - $\wedge$ conjunction sign
 - $\vee$ disjunction sign
 - $\supset$ implication sign
 - $\equiv$ sign of material equivalence
- the quantifiers:
 - $\exists$ existential quantifier
 - $\forall$ universal quantifier
- the equal sign: $=$
- the parentheses: (and)
- the separator sign: ,

The first-order language – non-logical constants

The symbols of the Con set can be divided into two parts.

$$Con = \bigcup_{i=0}^{\infty} (\mathcal{P}(i) \cup \mathcal{F}(i));$$

1. symbols used as name of functions:

- $\mathcal{F}(0)$ contains the name parameters,
- $\mathcal{F}(k)$ where $k \geq 1$ contains the function parameters.

2. symbols used as name of predicates:

- $\mathcal{P}(0)$ contains the propositional parameters,
- $\mathcal{P}(k)$ where $k \geq 1$ contains the predicate parameters.

The argument k in $\mathcal{F}(k)$ and $\mathcal{P}(k)$ describes the arity (number of arguments) of the function and predicate parameters.

First-order terms

The set of *Term* is defined inductively:

1. $Var \subseteq Term$;
2. $\mathcal{F}(0) \subseteq Term$;
3. if $f \in \mathcal{F}(k)$ for some $k \geq 1$ and $t_i \in Term$ for all $i = 1, \dots, k$, then

$$f(t_1, t_2, \dots, t_k) \in Term.$$

First-order formula (1)

The set of *Form* is defined inductively:

Atomic formula

1. $\mathcal{P}(0) \subseteq \text{Form}$;
2. $(t_1 = t_2) \in \text{Form}$ if
 - t_1 and t_2 are terms ($t_1 \in \text{Term}$ and $t_2 \in \text{Term}$)
3. $f(t_1, t_2, \dots, t_k) \in \text{Form}$ if
 - $f \in \mathcal{P}(k)$ for some $k \geq 1$ and
 - $t_i \in \text{Term}$ for all $i = 1, \dots, k$.

First order formula (2)

Non atomic (compound) formula

4. $\neg A \in \text{Form}$ if A is a formula ($A \in \text{Form}$);
5. $(A \circ B) \in \text{Form}$ if
 - A and B are formulas ($A \in \text{Form}, B \in \text{Form}$) and
 - $\circ$ is a binary connective ($\circ \in \{\wedge, \vee, \supset, \equiv\}$);
6. $Qx A \in \text{Form}$ if
 - Q is a quantifier ($Q \in \{\exists, \forall\}$),
 - x is a variable ($x \in \text{Var}$) and
 - A is a formula ($A \in \text{Form}$).

Direct subterm and set of subterms

The *direct subterm* of the term is defined as follows:

- if $t \in \text{Var}$ then it has no direct subterm,
- if $t \in \mathcal{F}(0)$ then it has no direct subterm,
- if $f(t_1, t_2, \dots, t_k) \in \text{Term}$ then the direct subterms are $t_1, t_2, \dots, t_k$.

The *set of subterm* of a term $t \in \text{Term}$ is a set of terms denoted by $ST(t)$ and is defined inductively as follows:

1. $t \in ST(t)$;
2. if $t' \in ST(t)$ and t'' is a direct subterm of t' , then $t'' \in ST(t)$.

Note that ST is a function such that:

$$ST : \text{Term} \rightarrow 2^{\text{Term}} \setminus \{\emptyset\}$$

Direct subformula, set of subformulae

The *direct subformula* of a formula is defined as follows:

- If $A \in \text{Con}$ then the A atomic formula has no direct subformula;
- if $B \in \text{Form}$ and $C \in \text{Form}$ are arbitrary formulas and $x \in \text{Var}$, then
 - the formula $\neg B$, $\exists x B$ and $\forall x B$ have one direct subformula: the formula B ;
 - the formulae $(B \wedge C)$, $(B \vee C)$, $(B \supset C)$ and $(B \equiv C)$ have two direct subformulae: the formula B and the formula C .

The *set of subformulae* of a formula $A \in \text{Form}$ is set of formulae denoted by $SF(A)$ and is defined inductively as follows:

1. $A \in SF(A)$;
2. if $B \in SF(A)$ and C is a direct subformula of B , then $C \in SF(A)$.

Note that SF is a function such that:

$$SF : \text{Form} \rightarrow 2^{\text{Form}} \setminus \{\emptyset\}$$

Construction tree

The construction tree of a formula $A \in \text{From}$ is the finite ordered binary tree defined as follows:

- each nodes are labelled by a formula,
- their root is labelled by the formula A ,
- a nodes labelled by $\neg B$, $\exists x A$ and $\forall x B$ has only one child labelled by B ,
- a node labelled by $(B \wedge C)$, $(B \vee C)$, $(B \supset C)$ or $(B \equiv C)$ has exactly two children labelled by B and C ,
- the leaves labelled by atomic formulae.

The construction tree of a term $t \in \text{Term}$ is the finite ordered tree defined as follows:

- each nodes are labelled by a term,
- their root is labelled by the term t ,
- a nodes labelled by $f(t_1, t_2, \dots, t_k)$ has exactly k children labelled by t_1 , t_2 and so on until t_k ,
- the leaves labelled by atomic terms.

Logical degree

The *logical degree* of a formula is a nonnegative integer defined as follows:

$$\ell : \text{Form} \rightarrow \{0, 1, 2, \dots\}$$

If $A \in \text{Con}$ then

- the logical degree of an atomic formula A is 0
 $\ell(A) \Rightarrow 0$;

if $B \in \text{Form}$ and $C \in \text{Form}$ are arbitrary formulas,

- $\ell(\neg B) \Rightarrow 1 + \ell(B)$;
- $\ell(B \circ C) \Rightarrow 1 + \ell(B) + \ell(C)$ where $\circ \in \{\wedge, \vee, \supset, \equiv\}$;
- $\ell(Qx B) \Rightarrow 1 + \ell(B)$ where $Q \in \{\exists, \forall\}$ and $x \in \text{Var}$.

Functional degree

The *functional degree* of a term is a nonnegative integer defined as follows:

$$\tilde{\ell} : \textit{Term} \rightarrow \{0, 1, 2, \dots\}$$

- If $t \in \mathcal{F}(0) \cup \textit{Var}$ then

$$\tilde{\ell}(t) \Rightarrow 0$$

(the functional degree of an atomic term t is 0)

- if $f(t_1, t_2, \dots, t_k) \in \textit{Term}$ then

$$\tilde{\ell}(t) \Rightarrow 1 + \sum_{i=1}^k \tilde{\ell}(t_i)$$

(the functional degree of a compound term is 1 plus the sum of the functional degrees of the direct subterms)

Scope of a connective

For a connective $\neg, \wedge, \vee, \supset, \equiv, \exists$ or $\forall$ appearing in a non-atomic formula A , the *scope* is the formula

- with the smallest logical degree

such that

- it is a subformula of A , and
- it contains the connective.

The *main logical connective* of a formula is the connective whose scope equals the formula itself.